

Photometric data based on 1000 lamp lumens

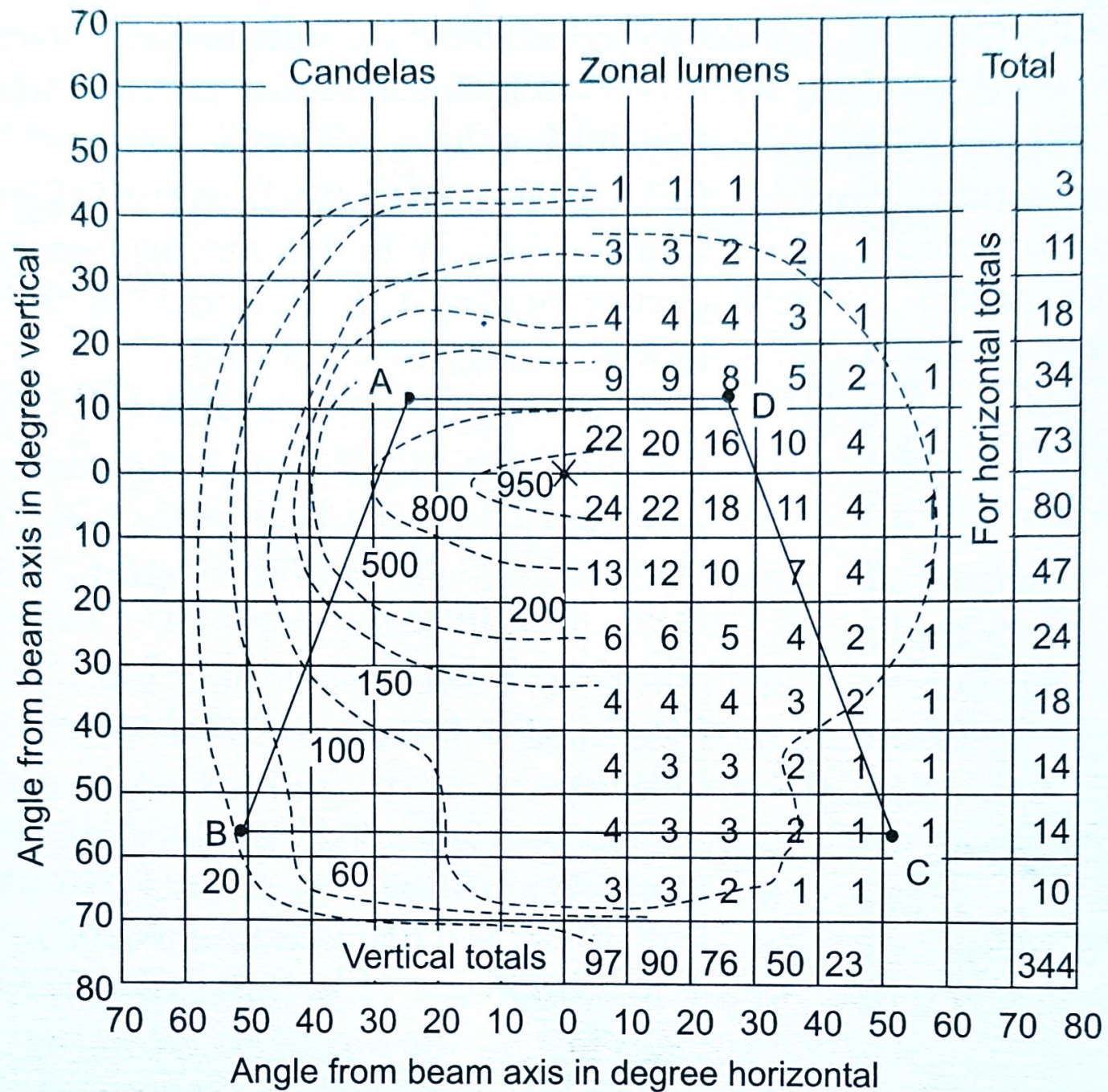


Table 11.3 Utilization factors for a bare fluorescent tube fitting with two 58 W 1500 mm lamps (%)

<i>Luminance factors</i>		<i>Room index</i>								
<i>Ceiling</i>	<i>Walls</i>	<i>0.75</i>	<i>1</i>	<i>1.25</i>	<i>1.5</i>	<i>2</i>	<i>2.5</i>	<i>3</i>	<i>4</i>	<i>5</i>
70	50	48	53	59	64	71	75	79	83	86
70	30	40	46	51	57	64	69	73	78	82
70	10	35	40	46	51	59	64	68	74	78
50	50	43	48	52	57	63	67	70	74	76
50	30	37	41	46	51	57	62	65	70	73
50	10	33	37	42	46	53	58	61	67	70
30	50	39	42	46	50	55	59	61	65	67
30	30	34	37	42	46	51	55	58	62	65
30	10	30	33	38	42	48	52	55	59	62

Table 11.2 Luminance factors for painted surfaces

Surface	Typical colour	Luminance factor range (%)
Ceiling	White, cream	70–80
Ceiling	Sky blue	50–60
Ceiling	Light brown	20–30
Walls	Light stone	50–60
Walls	Dark grey	20–30
Walls	Black	10
Floor	—	10

A drawing office $16\text{ m} \times 11\text{ m}$ and 3 m high has a white ceiling and light-coloured walls. The working plane is 0.85 m above the floor. New Streamlite double-lamp luminaires are to be used and their normal spacing-to-height ratio SHR is 1.75. Calculate the number of luminaires needed and draw their layout arrangement. Find the electrical power consumption of the lighting system.

